

Essential Concepts  
for  
Causal Inference  
in  
Randomized Experiments  
and  
Observational Studies

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# Prologue to Causal Inference

- My Introduction
  - Physics – Wheeler 1961
  - Experimental Design – Cochran 1968
- Clear Separation Between
  - Science = object of inference: DEFINE QUESTION FIRST
  - What is done to learn about the science
    - Intervene to measure aspects at a point in time
- Same notation/representation of science no matter how we try to learn about or measure it
- Missing data always exist
  - Cannot go back in time
  - Heisenberg uncertainty principle – observer effect

# Fisher (1925) – Actually Randomize

- Thereby create balance on covariates in expectation
- Rerandomize if assignment bad
  - Recondite advice
- Hypothetically rerandomize for assessment
  - Stochastic proof by contradiction of Fisher null hypothesis
  - Special case of posterior predictive p-value (Rubin, Annals 1984)
- If we say, “This boy has grown tall because he has been well fed,” we are not merely tracing out the cause and effect in the individual instance; we are suggesting that he might quite probably have been worse fed, and that in this case he would have been shorter (Fisher, 1918).
- But NO notation for concepts
- Geometry under the Fisher null hypothesis implies tests

# Neyman (1923) – Potential Outcomes

- Define estimands in randomized experiments as functions of Potential Outcomes – explicit notation
  - $Y(0)$  = Array of potential outcomes under treatment 0
  - $Y(1)$  = Array of potential outcomes under treatment 1
  - Etc., implicitly assuming SUTVA (Rubin 1978, 1980)
  - Cannot observe both on any one unit – “Heisenberg”
- Evaluate operating characteristics of procedures over randomization distribution
- Role of non-additive unit-level causal effects
  - non-zero correlation between potential outcomes – not directly estimable, but alters finite population intervals

# RCM (1974, 1975, 1977, 1978)

- Potential outcomes define causal estimands in all situations, not just in randomized experiments
  - Neyman disagreed (“too speculative”)
- Assignment mechanism *needed* for causal inference:  
Assignment of treatment indicators  $W$  given science

$$\Pr(W | X, Y(0), Y(1))$$

$$\text{Uncounfounded} = \Pr(W | X)$$

$$\text{Ignorable} = \Pr(W | X, Y_{\text{obs}}), \text{ as in sequential designs}$$

$$Y_{\text{obs}} = \text{observed values of } Y(0) \text{ and } Y(1)$$

- Bayes = model the science (1975, 1978)  
 $(X, Y(0), Y(1))$

# Potential Outcomes Approach to Causal Inference – Simplest Setting Under SUTVA

$W = 1 \Rightarrow$  active treatment (e.g., active treatment)

$W = 0 \Rightarrow$  control treatment (e.g., control treatment)

|   | $Y(1)$ | $Y(0)$ |
|---|--------|--------|
| 1 |        |        |
| . |        |        |
| . |        |        |
| . |        |        |
| N |        |        |

$Y(1)$  = outcomes if exposed to active treatment

$Y(0)$  = outcomes if exposed to control treatment

$\text{Ave}[Y_i(1) - Y_i(0)]$  = Average causal effect

$\text{Med}[Y_i(1) - Y_i(0)]$  = Median causal effect

# Fundamental Problem Facing Causal Inference

- For each  $i$ , only  $Y_i(1)$  or  $Y_i(0)$  can be observed

|       |   | $Y(1)$ | $Y(0)$ | $W$ |
|-------|---|--------|--------|-----|
| Units | 1 | ✓      | ?      | 1   |
|       | . | ✓      | ?      | 1   |
|       | . | ✓      | ?      | 1   |
|       | . | ?      | ✓      | 0   |
|       | . | ?      | ✓      | 0   |
|       | N | ?      | ✓      | 0   |

- Random assignment of active versus control representative sample of  $Y_i(1)$  will be compared to representative sample of  $Y_i(0)$  – obvious?
- Peirce, Neyman, Gosset, Fisher,
- Under Fisher null,  $Y_i(1)=Y_i(0)$  for all  $i$

# “Mistake” of OLS and Companions

- Lost potential outcomes and key Fisher/Neyman concepts – observed value notation

$$Y_{\text{obs},i} = W_i Y_i(1) + (1 - W_i) Y_i(0)$$

- Mixes assignment mechanism and science – BAD!!!
- Suppresses key insights – retreat to confusion
- Standard in biostatistics, economics, epidemiology, everywhere!!
- Even great statisticians (e.g., Fisher, Cochran, Cornfield, ...) confused themselves using OLS with  $Y_{i,\text{obs}}$  notation



# Hunter's Paradox

- Sloppier statistical thinking could lead to more success than misapplied “careful” thinking
- Suppose going north to hunt was more successful than going south: 60% vs. 40%
- Careful “data mining” tribe noticed this and went north 60% and south 40%
- Statistically naïve tribe ignored this information and went north 100%
- Which tribe ate better?

# Why is OLS style so Bad?

- Buried crucial issues – role of additivity
- Inherent missing data issue lost using just  $Y_{i,obs}$
- No longer any need to impute  $Y_{i,mis}$
- Estimands must be parameters in model
- Asymptotic silliness and sloppy mathematics
  - E.g., distinction between finite and super pop
- Emphasis on *analysis and estimators* rather than *study design and estimands*

# Why was OLS style so dominant?

- Computational limitations in the past
  - No Fisher randomization tests possible in general
  - Not possible to simulate Neyman operating characteristics of procedures in small samples or under various alternative assumptions
  - Using “mindless off the shelf” batch programs became the standard and therefore acceptable in publications
- People who understood issues in randomized experiments tended to avoid observational studies, and that generation largely passed
  - Do not fall into this trap!!

# Terrible Influence on the Way Observational Studies Were Treated

- Association versus causation -- muddled
- Case – noncase studies

$Y_{i,obs}$ ,  $W_i$  both dichotomous, so logistic regression  
“works” either way -- what is  $Y_{i,mis}$ ?

Assignment Mechanism }  
Sampling Mechanism } Both lost

*“We now consider the distinction between the kinds of inferences that can be supported by observational studies, whether prospective or retrospective, and those that can be supported by experimental studies. That there is a distinction seems undeniable, but its exact nature is elusive.” Cornfield, 1959, Principles of Research*

# Design Observational Studies to Approximate Randomized Trials

1. Hide outcome data until the design phase is complete
  2. Think very carefully about decision makers and the key covariates that were used to make treatment decisions
  3. If key covariates are not observed or very noisy, usually best to give up and seek better data source
  4. Find subgroups (subclasses or matched pairs) in which the treatment and control groups have balance – essentially the same distribution of observed covariates
    - Not always possible to achieve balance
    - Inferences are limited to subgroups where balance is achieved
  5. Protocol specified analysis
- #1 - #5 combine to create an objective design that approximates a randomized trial in each subclass that is balanced with respect to observed covariates

# Key -- Estimated Propensity Scores

- Diagnostics for checking balance are critical
- Objective because outcome-free design
- Use modern computation to aid visualizations of balance -- e.g., (Thomas) Love plots, then “drill down” using more refined tools
- Write protocol for analyses with buy-in from all stakeholders
- Anticipates “design trumps analysis” lecture

## When Conducting Causal Inference, Must Remember Lessons From Randomized Experiments

- Create balance in key covariates by rejecting, at design stage, bad randomizations
  - Goal in an observational study is to reconstruct a balanced randomized experiment, e.g., by blocking
  - If a completely randomized experiment, were the goal, then want 5% of covariates to “significantly” differ in treatment and control groups!
  - So understanding design of balanced experiments a crucial bridge to the design of observational studies